

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	XXXXXX
Project Name	Smart-Lender-Loan-Eligibility-Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python, HTML, CSS and JavaScript files follow proper indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into dataset, TRAINING, flask, templates, static, and model files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as applicant details, loan amount, credit history, prediction result, etc., are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate preprocessing, prediction, and application flow.
5	Code Comments	Inline and block comments explain log	Yes	Important sections are documented to improve understanding and maintenance.
6	Modular Design	Code is split into reusable functions/modules	Yes	Training, preprocessing, Flask application, and frontend are separated into different modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	The implementation avoids unnecessary duplicate code and keeps modules concise.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic validation and error handling are implemented to manage invalid user input.

Code Layout Checklist:

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles all prediction requests	High
2	Loan Eligibility Prediction Model (model.pkl) Data Preprocessing	Scikit-learn	Used for every loan prediction	High
3	Module HTML Templates	Pandas, NumPy	Used before every prediction	High
4		HTML5	User Interface	Medium
5	Static Resources	CSS3, JavaScript	UI styling and interaction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for training, deployment, templates, and static resources.
Readability	5	Code is easy to understand with meaningful names and proper formatting.
Reusability	5	Machine learning model, preprocessing logic, and Flask application can be reused or extended easily.
Documentation / Comments	5	Code is documented sufficiently for understanding and future maintenance.
Overall Score	5 / 5	The project follows a modular, maintainable, and reusable architecture suitable for deployment and future enhancement.